

User Guide: Data Collection & Visual Mapping

Distribution Analysis & Navigation Algorithm (D.A.N.A.) Framework

Project Creator:	Dana Makinen	Supported Metrics:	Imperial (Inches) / Metric (Centimeters)
Downstream Tool:	Library Space Management Tool	Component:	Data Collection Workbook

Part 1: Pre-Collection Spreadsheet Setup (Project Lead)

Before deploying staff to take physical measurements, Project Leads must configure the Data Collection Workbook to ensure accurate automated calculations and a seamless workflow. If any part of this configuration or layout seems unclear, it is highly recommended to reference the baseline setup in the companion file: **"Example Data Collection Workbook - 11th Floor 2026"**.

1.1 Understanding & Configuring the "Average Shelf Length"

Library ranges frequently feature varying shelf physical dimensions. To safely accommodate these variations without disrupting capacity algorithms, the framework tracks spatial volume via a standardized average shelf size computed per range side. The exact parameter value is determined by dividing the cumulative maximum width capacity of all shelves on a specific side by the absolute count of shelves on that side.

$$\text{Average Shelf Length} = \text{Total Combined Maximum Width of Shelves} / \text{Total Number of Shelves}$$

Practical Example: If a range side contains 66 shelves that hold a maximum of 35.5" each, and 11 base shelves that hold a maximum of 35.875" each, the calculation behaves as follows:

- 66 shelves × 35.5" = 2343"
- 11 shelves × 35.875" = 394.625"
- Total Capacity = 2737.625" ÷ 77 Total Shelves = 35.5535" (Recorded on sheet as 35.55)

Critical Data Placement: Project leads must input this calculated metric into the **Range Summary** sheet under the "Average Shelf Length For Each Range" column (Column H) for its corresponding range side.

1.2 Labeling Range Sections & Handling Architectural Obstructions

Project Leads must label all sections located on each applicable Range # Sheet tab before measurement begins:

- **Standard Naming Format:** Section markers must be systematically indexed to mirror logical shelf rows. For example, Range 1 sections should be labeled 1-1 through 1-9 on the first face, continuing from 1-10 through 1-18 on the reverse face. Range 2 follows the same convention (2-1 to 2-9, and 2-10 to 2-18).
- **Handling Structural Columns & Obstructions:** If a concrete column or structural pillar breaks the continuity of a physical section, label that section header explicitly as "Pillar". Leave the primary measurement grid boxes directly underneath empty, untouched, and gray.

LEAD BEST PRACTICE: LOCK SHEET PROTECTION

Once section headers and architectural markers are fully inputted, it is highly recommended that Project Leads lock down all formula logic cells and configuration areas, exposing only the raw physical measurement entry grids to field personnel to prevent accidental overwrites:

- **Google Sheets:** Use the **Protect sheets and ranges** tool (Data → Protect sheets and ranges) to set permissions for specific cell ranges or entire sheets.
- **Microsoft Excel:** Unlock the measurement entry cells (Format Cells → Protection → uncheck *Locked*), then protect the worksheet via **Review** → **Protect Sheet**.

Part 2: Field Measurement & Data Entry Protocol (Staff)

Once the workbook setup has been validated by the Lead, data collection staff may begin tracking live physical inventory metrics directly inside the designated grids.

LIBRARY LAYOUT CUSTOMIZATION NOTE

If the data collection sheets do not perfectly fit your physical library shelf layout, you will need to make localized adjustments to the spreadsheet by inserting columns or shelves accordingly to mirror your environment.

2.1 Adaptive UI Color Transitions

The collection grids utilize a dynamic user interface designed to scale natively to any range height. When field data is introduced into the originally grayed-out entry rows, background colors instantly morph to track operational state:

- **Gray Shading:** Signifies an empty, unrecorded shelf row.
- **White Shading:** Signifies an actively filled cell, visually confirming data capture.

2.2 Numeric Format and Rounding Requirements

NO FRACTIONS ALLOWED

Fractional values (e.g., "1/2" or "3/4") are completely unsupported by the framework. Entering plain text fractions breaks the downstream parsing scripts. All measurements must be input exclusively as numeric decimals.

Data collectors should round metrics to the nearest whole inch, half-inch, or quarter-inch depending on the precision rules established by the Lead for speed and consistency:

- **Nearest Whole Inch:** Logged as integers (e.g., 12, 24).
- **Nearest Half Inch:** Logged as .5 decimal increments (e.g., 12.5, 24.5).
- **Nearest Quarter Inch:** Logged as .25 or .75 decimal increments (e.g., 12.25, 24.75).

UNIT MEASUREMENT FLEXIBILITY

Although baseline templates demonstrate measurements using imperial inches, the workbook calculations adapt seamlessly to metric setups. If your library utilizes centimeters, ensure that capacity values are logged uniformly throughout every workbook tab.

Part 3: Dynamic Heat Map Generation

Upon completing field data log passes and populating the initial layout parameters inside the 'Floor_Blueprint' tab, the system will instantly construct a top-down visual representation of collection load levels.

CRITICAL BLUEPRINTING RESTRICTION

It is not advised to copy, paste, or enter any text directly into the cells of the **Heat Map** sheet tab. Active lookup and calculation formulas reside inside every single cell of that tab. To map out your floor grid overview, you must exclusively use cell borders rather than content input.

3.1 Blueprinting the Layout Grid

Navigate directly to the **Floor_Blueprint** tab. The coordinate canvas will load blank. Replicate your structural floor plan layout from a bird's-eye perspective using cell border strokes.

1. **Step 1:** Enter the initial section index code (e.g., 1-1) into the cell corresponding to its literal room position.
2. **Step 2:** Select the cell and drag its bottom-right corner autofill handle outward to automatically project the sequence out to 1-9.
3. **Step 3:** Perform the identical sequence on the opposite face to generate markers 1-10 through 1-18.
4. **Step 4 (Structural Obstructions):** If a structural pillar breaks up a row, leave that relative coordinate cell completely blank in your blueprint layout to preserve correct spatial rendering.

3.2 Reading the Interactive "Heat Map" Sheet

The Heat Map view visually mirrors the structural cells built in your blueprint canvas. Rather than presenting raw numeric values, it automatically computes space utilization metrics and tints cell fills via a high-contrast, colorblind-friendly alert scheme:

- **Red:** 80% or higher capacity full. Critical density threshold breached. Requires shift intervention.
- **Yellow:** 70% capacity full. Standard warning threshold. Nearing maximum threshold.
- **Blue:** 60% or lower capacity full. Healthy headroom. Substantial expansion capacity remains available.

Part 4: Data Export for Shifting Tool Integration

The central Range Summary tab automatically aggregates field statistics across all range profiles into an aligned schema table. This summary matrix functions directly as the import source file for your downstream Python-driven Library Space Management Tool.

Export Workflow Steps:

1. Navigate to the completed **Range Summary** tab inside the active workbook.
2. Export the tab as a CSV file:
 - **Google Sheets:** Go to **File** → **Download** → **Comma Separated Values (.csv)**.
 - **Microsoft Excel:** Go to **File** → **Save As**, select your target folder, and select **CSV (Comma delimited) (*.csv)** from the file type options.
3. Save the resulting text file to your local computer. It is now formatted perfectly to load straight into the D.A.N.A. shifting optimizer scripts.